

Document 30. Worked Essay Exemplars (6 annotated A-grade essays)

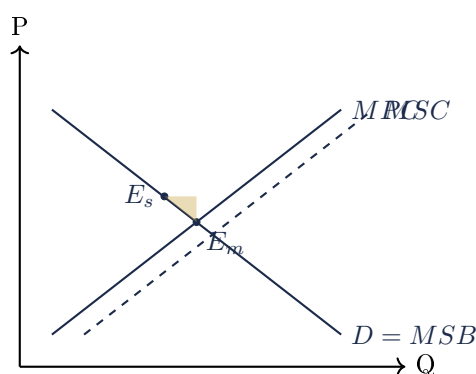
Six full essay exemplars with marginal commentary. The marginal notes (in gold italics) flag the structural moves, evaluation points, and context-anchoring that earn Level 3 marks.

Section A. Exemplar 1 — Externalities and Singapore's Carbon Tax

Question: (a) Explain how a negative externality leads to market failure. [10] (b) Discuss whether the carbon tax is the most appropriate policy to address the negative externality of carbon emissions in Singapore. [15]

(a) A negative externality is a spillover cost imposed on third parties not party to the transaction. Carbon emissions from electricity generation impose costs on others through poorer air quality, climate change, and adverse health outcomes. *[Definition; context anchor in opening]*

Free-market producers maximise private profit at $MPC = MPB$, yielding equilibrium E_m at quantity Q_m and price P_m . However, since social cost $MSC = MPC + MEC$ exceeds private cost, the true social optimum lies at E_s where $MSC = MSB$, with quantity $Q_s < Q_m$. *[Mechanism; key equation]*



For every unit between Q_s and Q_m , $MSC > MSB$. The triangle bounded by MSC, MSB and the vertical line at Q_m represents the deadweight welfare loss to society. The market thus over-produces relative to the social optimum, allocating too many resources to carbon-emitting activities. *[Diagram + welfare analysis with DWL identified]*

In sum, the price mechanism fails because it does not internalise the external cost, leading to allocative inefficiency. *[Concluding link]*

(b) Singapore introduced a carbon tax in 2019 at S\$5/tCO₂e, raised to S\$25 in 2024, and slated to reach S\$45 by 2026–27 and S\$50–80 by 2030. The tax targets facilities emitting $\geq 25,000$ tCO₂e per year. *[Strong context anchor with data]*

In favour of the carbon tax, it is a Pigouvian policy that internalises the externality by raising MPC up to MSC. New private equilibrium coincides with E_s , eliminating the DWL. Revenue is recycled into clean-energy R&D, which over time shifts the SRAS curve rightward (technology subsidies), achieving sustainability without sacrificing long-run growth. The market-based design preserves firms' flexibility to choose the cheapest abatement route, ensuring productive efficiency. *[One-sided analysis with mechanism]*

However, the current rate may be below the true MEC. The Inter-American Development Bank and IPCC estimates put the social cost of carbon at US\$50–150 per tCO₂e — significantly above Singapore's S\$25 rate. The tax also exempts small emitters and households, leaving the bulk of consumption emissions untouched. *[Magnitude evaluation; data citation]*

There is also a regressive distributional concern: utility prices have risen for low-income households. Singapore mitigates this via U-Save rebates and the Assurance Package, but the targeting is imperfect. *[Distributional move]*

Alternative policies include tradable permits, regulation (emission standards), direct provision (solar grants), and education campaigns. A tradable permit scheme would set a hard cap and allow market-determined prices, ensuring environmental certainty; however, cap-setting is politically contentious. Regulation is more direct but creates no incentive for over-compliance. *[Alternative policies; C-E-F applied]*

On balance, the carbon tax is the most appropriate single policy because (i) it preserves market efficiency, (ii) it generates revenue that funds complementary supply-side investments, and (iii) it sends a clear price signal that scales with the escalator. However, no single policy is sufficient. Singapore's actual approach pairs the tax with the Green Plan 2030, EV adoption schemes (VES, EEAI), and binding solar generation targets. *[Stance; synthesis with policy mix]*

In conclusion, the carbon tax is the appropriate *anchor* of climate policy, but its effectiveness depends on the escalator reaching a level closer to the social cost of carbon by 2030, paired with regulation in non-priced sectors and targeted transfers for low-income households. *[Conclusion with forward-looking judgement]*

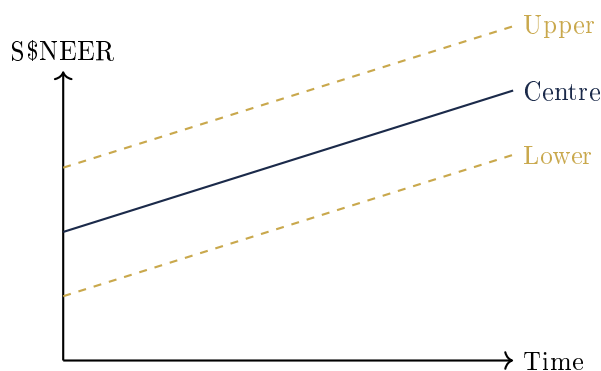
Section B. Exemplar 2 — MAS Monetary Policy

Question: (a) Explain how monetary policy works in the Singapore context. [10] (b) Evaluate the effectiveness of MAS's exchange-rate-centred framework in achieving price stability. [15]

(a) The Monetary Authority of Singapore (MAS) operates an exchange-rate-centred monetary policy, targeting the Singapore Dollar Nominal Effective Exchange Rate (S\$NEER) rather than an interest rate. *[Define with context immediately]*

Singapore is a small open economy with import content of roughly 40% of expenditure. The exchange rate has a large pass-through to domestic prices and to net exports. By managing the S\$NEER, MAS directly controls imported inflation through the SRAS curve, and influences aggregate demand through $(X - M)$ in the AD curve. *[Economic structure justifies tool]*

The framework operates within a Basket-Band-Crawl (BBC) policy band, with three policy levers: slope (rate of appreciation), width (band tolerance), centre (re-centring up or down). To tighten, MAS steepens the slope and/or re-centres upward. To ease, MAS flattens the slope, widens the band, or re-centres downward. *[Mechanism with policy levers]*



The impossible trinity forces Singapore to give up independent interest-rate policy: domestic SORA tracks US Fed rates. This is acceptable because domestic consumption and investment are relatively interest-insensitive given low household leverage and the dominance of MNCs in the corporate sector. *[Impossible trinity link]*

(b) Effectiveness: MAS has used the exchange rate tool five times to tighten over 2021–23 in response to imported and core inflation. By Q4 2024, core inflation had moderated from a peak of 5.5% to ~2%, within the implicit medium-term target. *[Empirical evidence; magnitude]*

Strengths. (i) Direct on imported inflation: a stronger S\$ lowers the SGD cost of imports, shifting SRAS rightward. (ii) Anti-inflationary via AD: appreciation reduces X and raises M , dampening AD. (iii) Insulates Singapore from US-driven imported inflation, an advantage when the Fed is on a different cycle.

Limitations. (i) Cannot address purely domestic cost-push (e.g. wage-price spiral) without supply-side complements. (ii) Hurts export competitiveness during sustained tightening; non-tradable sectors absorb the brunt. (iii) The interest rate is determined externally, leaving Singapore unable to use rate policy for domestic credit cycles. (iv) Sustained appreciation may invite trade friction with partners. *[Multi-channel evaluation]*

Comparison to interest-rate framework. If Singapore adopted a Fed-style rate-targeting regime, capital flows would arbitrage any deviation from US rates, forcing either capital controls or a fixed peg. Neither is consistent with Singapore's role as a global financial hub. *[Alternative policy analysis]*

On balance, the MAS framework is highly effective for price stability in a small open economy. The 2022–23 tightening preserved Singapore's price stability while peer economies experienced higher and more prolonged inflation. *[Stance with comparative evidence]*

In conclusion, the framework is well-calibrated for Singapore's structural openness, but it cannot be expected to handle every shock alone. Supply-side investment (productivity, food security, energy diversification) and targeted fiscal transfers (U-Save, Assurance Package) form the complementary pillars of a coherent policy mix. *[Conclusion with policy-mix synthesis]*

Section C. Exemplar 3 — Free Trade vs Protectionism

Question: (a) Explain the law of comparative advantage. [10] (b) Discuss whether protectionism can ever be justified in a small open economy like Singapore. [15]

(a) Comparative advantage states that a country should specialise in producing the good for which its opportunity cost is lowest. Even if one country has absolute advantage in both goods, both gain from trade as long as opportunity costs differ. *[Definition; key insight]*

Suppose Country A produces 4 cars or 8 textiles per worker-hour, while Country B produces 1 car or 4 textiles. A's opportunity cost of 1 car is 2 textiles; B's is 4 textiles. Therefore A has comparative advantage in cars. Conversely, B's opportunity cost of 1 textile is 0.25 cars; A's is 0.5 cars. Therefore B has comparative advantage in textiles. *[Numerical illustration anchors the abstract concept]*

At any terms of trade between 2 and 4 textiles per car, both countries can consume *outside* their own PPCs, yielding gains-from-trade. Static gains include lower prices and wider variety; dynamic gains include technology transfer and scale economies. *[Multiple sources of gains]*

(b) *The case for protectionism.* (i) *Infant industry:* temporary protection allows a nascent sector to attain scale, after which protection is removed. (ii) *Strategic / national security:* defence, semiconductors, food. (iii) *Anti-dumping:* prevent foreign predatory pricing. (iv) *Sunset industries:* cushion job losses during structural transition. *[Standard 4-fold case]*

Application to Singapore. (i) Singapore has no infant industries by virtue of size; most production is at the technology frontier. (ii) Strategic concerns are real — 90% of food is imported, energy is fully imported. (iii) Anti-dumping is rare given Singapore's WTO commitments and active dispute participation. (iv) Sunset industries are addressed by retraining (SkillsFuture, CCPs), not protection. *[Test each argument in SG context]*

The case against protectionism in Singapore. (i) Tariffs raise consumer prices and producer input costs, both of which damage Singapore's hub competitiveness. (ii) Retaliation under FTA agreements is near-certain given Singapore's 27+ FTAs. (iii) Welfare loss from two DWL triangles plus dynamic loss from reduced competition. (iv) Singapore's success depends on its rules-based, open identity. *[Quantified arguments]*

However, strategic resilience can use non-tariff instruments. (i) *Diversification:* 30-by-30 food production target; multi-source energy procurement. (ii) *Stockpiling:* National Stockpile for essential goods. (iii) *Trade agreements:* deepen FTAs to reduce supply chain risk. *[Distinguish protectionism from resilience]*

On balance, classical protectionism is not justified in Singapore. The infant-industry, anti-dumping, and sunset-industry arguments do not apply with sufficient force, while the costs (consumer welfare loss, retaliation, reputational damage) are large. However, *strategic resilience policies* that achieve the security goal without restricting trade are appropriate, and Singapore has pursued these via diversification, stockpiling, and FTA deepening. *[Carefully distinguished conclusion]*

In conclusion, protectionism is not the right tool for Singapore. The proper response to deglobalisation is selective engagement via FTAs (RCEP, CPTPP), supply-chain diversification, and human capital investment, not tariff walls. *[Forward-looking conclusion]*